

Select, Compress, Reinvest: A Controlled Study of Visual-Token Allocation in Long-Video MLLMs

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Abstract

An hour of video sampled once per second is 3,600 frames; a video language model can usually afford eight. Choosing those eight well matters more than the field’s default suggests: on hour-long videos, eight query-selected frames answer more questions correctly than sixteen uniformly spaced ones. Establishing that comparison is hard, because published selectors change the frame scorer, the prompt, the resolution policy, and the answering model together. We hold all four fixed and vary one decision at a time. Under a single frozen LongCLIP scorer that sees only the question stem, we compare six training-free selection rules at eight frames on LongVideoBench, VideoMME, and LVBench. Query-aware selection is the dominant lever and needs no new algorithm: classical Orthogonal Matching Pursuit (Pati et al., 1993), published in 1993, beats uniform sampling by 5.7, 5.9, and 11.8 points and stays within one point of a purpose-built modern selector. That ordering survives a swap to SigLIP, within a test that bounds only large scorer effects. Holding timestamps fixed, halving each frame’s spatial budget moves accuracy by at most 0.44 points; spending the savings on more frames adds 2.36 points on LongVideoBench’s long bins at verified equal cost. Select relevant frames first, compress them second.

1 Introduction

A video language model cannot look at a whole video. Decoded at one frame per second, a ten-minute clip is 600 images and an hour-long one is 3,600; a typical long-video system encodes eight. Everything the model will ever know about the video passes through that handful of frames, so the rule that picks them is the first and tightest bottleneck in the pipeline, not a preprocessing detail.

The default rule is still uniform sampling, regardless of what was asked, and our clearest single result is that this default is expensive. On hour-long LongVideoBench videos, eight frames chosen for their relevance to the question beat sixteen uniformly spaced frames by 6.9 points ($p=.0011$), and the same pattern holds one budget lower on ten-minute videos. Half the frames, better answers: spending compute on which frames to keep buys more than spending it on how many.

Many systems already replace uniform sampling with query-aware retrieval, coverage, diversity, or learned evidence scores (Tang et al., 2025; Sun et al., 2025; Zhu et al., 2025b; Wang et al., 2026a; Peng et al., 2026), and others jointly adapt frame choice and resolution (Zhang et al., 2025a; Chen et al., 2026a; Wang et al., 2026b) or prune tokens inside the vision stack (Li et al., 2026; Hong et al., 2026). They report large end-to-end gains, but leave a narrower question open: holding the scorer, frame budget, and answering model constant, which selection rule supplies the most useful evidence? Published tables cannot answer it, because those components move together. A selector can look strong because it ships with a better encoder; a dynamic-resolution system can look strong because it fits in more frames rather than because it allocates resolution well. Comparing two numbers from two papers compares two experiments, not two ideas.

This paper isolates the pieces. We fix one frame scorer, one prompt boundary, one answering model, and one evaluation harness, then intervene on a single decision at a time. Select changes which timestamps are chosen at fixed count and resolution; compress holds those timestamps and shrinks each frame’s spatial budget; reinvest spends the recov-

ered budget on more timestamps rather than sharper ones. Each step is a paired comparison on the same questions, so its difference is attributable to that step alone. A fourth comparison swaps the scorer itself, checking that the resulting ordering is not an artifact of the encoder we happened to freeze.

The rule at the centre of these interventions is deliberately old. Orthogonal Matching Pursuit (Pati et al., 1993) is a greedy sparse-approximation algorithm from 1993: it repeatedly takes the candidate most correlated with what the query still needs, then projects that direction away before choosing again. We adopt it unmodified and untuned, which is a methodological choice rather than a concession. An off-the-shelf rule has no hyperparameters for us to fit to these benchmarks, so a gap between it and a competing selector reflects the controlled comparison rather than unequal effort on our side of it. It also sets an informative bar: how well the least sophisticated thing that could plausibly work does here measures how much recent purpose-built selectors gain from their subset rules as opposed to their scorers and pipelines.

The three interventions rank cleanly, and that ordering is the paper’s main practical output. Selection is the large lever, improving on uniform sampling by 5.7 to 11.8 points depending on the benchmark and beating uniform inputs that carry twice as many frames. Compression is nearly free but useless alone: halving the per-frame budget at fixed timestamps changes accuracy by at most 0.44 points, with formal ± 3 -point equivalence on pooled long videos. Reinvestment converts that free budget into two to three points. None of this is universal—the benefit depends on which model reads the frames and which benchmark asks the question, and OMP’s later picks drift toward content that is visually novel but irrelevant—and we report those boundaries rather than smoothing them.

Contributions.

- A matched comparison of six training-free selectors under one scorer, one prompt boundary, one frame budget, and one answerer harness. OMP is a strong query-focused rule and stays within one point of LDDR’s stage-1 selector.

- Evidence that selection substitutes for frame count. OMP significantly outperforms uniform sampling while using half as many frames, in two independent long-video cross-budget contrasts.
- A decomposition of the fixed-budget allocation decision. Roughly halving the per-frame spatial budget preserves accuracy, with formal equivalence testing rather than a bare null result; reinvesting the savings in additional keyframes then improves long-video QA.
- Evidence that the ranking survives a scorer swap. Replacing LongCLIP with SigLIP changes 67–84% of the selected frames yet leaves the selector ordering intact, within a test that bounds only scorer effects above roughly five points.
- Explicit boundaries on all three claims, from paired tests, two answerer families, three benchmarks, a residual-geometry diagnostic, and a qualitative failure audit that identifies when query-focused diversity helps and when it drifts off-topic.

2 Related Work

Query-aware frame selection. Training-free selectors combine relevance to the question, coverage of the timeline, and diversity within the chosen set, in different proportions: AKS recursively partitions the timeline for coverage (Tang et al., 2025), FOCUS treats clips as arms in a budgeted bandit (Zhu et al., 2025b), and MDP3 uses a DPP-based dynamic program (Sun et al., 2025), as do AdaRD-Key and Adaptive Greedy (Zhang et al., 2025b; Huang and Zhu, 2026). Segment-based methods make the structure explicit through per-segment quotas (Peng et al., 2026) or event-like segments refined by maximal marginal relevance (Chen et al., 2026b), while others add uncertainty-triggered computation (Kim et al., 2026) or learn frame-level conditional evidence (Wang et al., 2026a). These differ not only in the subset rule but in scorer, candidate pool, training requirements, and frame budget, so reported gaps between them confound the rule with its pipeline. Our comparison removes that confound by giving every rule the same cached scores.

Selection with spatial adaptation. A second line couples frame choice to resolution: Q-Frame pairs query-aware selection with multi-resolution scaling (Zhang et al., 2025a), LDDR combines a linear-DPP selector with Group-DPP importance and dynamic resolution (Chen et al., 2026a), and DAFS optimizes candidate-pool size and per-frame token budget from MLLM attention (Wang et al., 2026b). A joint gain has at least three possible sources—better timestamps, better resolution policy, or simply more temporal coverage—and none of these papers separates them, which is exactly what our select/compress/reinvest sequence does.

Visual-token reduction. Token-level methods cut cost during or after encoding, learning retain/compress/drop decisions (Hou et al., 2026), assigning group budgets from answerer uncertainty (Qi et al., 2026), pruning with query guidance (Li et al., 2026), or using scene structure and motion (Hong et al., 2026). Concurrent work poses our question directly: AdaAlloc (An and Grauman, 2026) asks how much of a fixed budget to spend on global context versus high-resolution local evidence; no preprint or code was public at the time of writing, so we describe it from its project-page abstract and draw no comparison against our numbers. We ask a coarser but separable question upstream of all of them—before any token-level pruning, does an answerer gain more from sharper frames or from additional moments?—and our target is the evidence such an allocator needs rather than the allocator itself.

3 Method: Three Controlled Interventions

3.1 The experimental unit

Every comparison in this paper is paired. The unit is a single benchmark question answered twice, once under each of two input policies, with everything else held constant. This lets us count the questions each policy wins rather than compare two accuracy numbers that may differ for unrelated reasons. The three interventions differ only in what the two policies are allowed to change: selection changes the timestamps while holding the frame count, scorer, and resolution fixed; compression holds

the timestamps fixed and changes the per-frame spatial budget; reinvestment spends the budget saved by compression on additional timestamps. Measuring the marginal value of one decision at a time is what distinguishes this from a comparison of complete systems whose components all differ at once. A fourth intervention sits outside that sequence and tests the design itself: replacing the scorer while holding the selection rule, budget, and answerer fixed asks whether the ordering the first three produce is a property of the rules or of one encoder (§5.2).

3.2 One scorer, shared by every rule

Videos are decoded at 1 fps. Each candidate frame and the question stem are encoded once with LongCLIP (Zhang et al., 2024), and the resulting embeddings and stem similarities are cached. Every selector then reads the same cache, so no rule benefits from a better encoder than another.

Answer options never enter selection. That is a convention rather than a neutral choice, so we measure what it costs. Scoring against the fused question-and-options string instead of the stem alone changes 41.9% of the frames top- k selects on LongVideoBench-600s and 53.3% of those OMP selects; OMP is the more sensitive of the two because each pick is chosen to explain a residual component of the query vector, so a perturbation of that vector compounds through the greedy chain. Those are also better frames. On that bin, with the fused query, OMP reaches .6699 against .6311 for the stem (+3.88 points, $p=.033$) and top- k reaches .6408 against .6068 (+3.40, $p=.087$). Those are better frames, but not because of the options: a control that swaps in an unrelated video’s options, matched on length, already reaches .6529—56% of the gain—so the effect is not specific to the answer set, and neither contrast separates at this sample size. We nonetheless score with the stem alone, following the official AKS implementation, which passes the question without candidates to its CLIP and BLIP branches (Tang et al., 2025). Every selector in this paper is therefore evaluated in a deliberately conservative configuration, and the gap above is the price of that choice rather than a hazard it avoids. The consequence for reading the literature is the

substantive one: the text handed to the scorer is an uncontrolled axis worth close to four accuracy points, so a comparison that is inconsistent about it is measuring the prompt rather than the selector.

3.3 Selection rules

The matched comparison contains uniform sampling, cosine top- k , AKS (Tang et al., 2025), FOCUS* (Zhu et al., 2025b), OMP, and LDDR-select (Chen et al., 2026a). Every rule here consumes our LongCLIP stem scores, but that is native only for LDDR-select. AKS and FOCUS both score with BLIP-ITM in their published form, so both are reproductions of a subset rule under a substituted scorer, and LDDR-select is the only row evaluated with the encoder its authors used—an advantage to keep in mind when reading its position in Table 1. FOCUS* carries a star for a second reason: it replays the published clip-bandit schedule against the dense LongCLIP vector rather than the original budgeted online ITM process, which is the substance of the method; Appendix B states exactly what this does and does not test. LDDR-select is the stage-1 Linear-DPP selector (LD) only, not the full dynamic-resolution pipeline; it is a published, separately tabulated configuration of that system rather than a truncation we invented.

Let $E \in \mathbb{R}^{N \times d}$ contain L2-normalized frame embeddings and let q_0 be the normalized stem embedding. At each step OMP takes the frame most correlated with the current residual, then subtracts the span of everything already selected from the query:

$$\begin{aligned} b_r &= \arg \max_{i \notin B_{r-1}} e_i^\top q_{r-1}, \\ q_r &= q_0 - \text{Proj}_{\text{span}\{e_j: j \in B_r\}}(q_0), \end{aligned} \quad (1)$$

where $B_r = B_{r-1} \cup \{b_r\}$. In its original signal-processing setting this is sparse reconstruction: the residual shrinks as the selected atoms explain more of the target. Whether that interpretation survives in a contrastive image-text embedding space is an empirical question, not an assumption, and §5.5 tests it directly. What the projection reliably does here is suppress directions already covered by earlier picks.

3.4 Spatial budget and reinvestment

For the fixed-timestamp intervention, frames are resized before reaching the model’s processor, whose own internal resize is disabled for Qwen3-VL so that our resize is the only one applied. The main compressed arm uses a residual-proportional schedule targeting a mean spatial fraction of about 0.53, which we label D@53.

Two controls test what the schedule’s shape contributes. A flat split at the same mean fraction asks whether OMP’s ranking is a useful priority for spending pixels. A reconstruction of LDDR’s stage-2 Group-DPP importance, applied to identical OMP-8 timestamps, asks whether a different importance signal can do better than either.

The reinvestment arm selects $k=16$ timestamps and applies an approximately 50% per-frame cap—a different quantity from the 0.53 mean used when timestamps stay fixed. Because the comparison only means something if the two arms really cost the same, we audit tokens empirically rather than assuming the resize ratio transfers: a script reproduces Qwen’s smart-resize rules from each video’s own dimensions. The sixteen-frame arm consumes 0.996 and 0.984 times the tokens of the eight-frame full-resolution arm in the 600s and 3600s LongVideoBench bins. Both ratios are below one, so the LongVideoBench reinvestment result is conservative: the winning arm is also the cheaper one. Video-MME and LVBench use the same resize rule, but we did not run a separate token audit on them and do not claim audited parity there.

4 Evaluation Setup

We evaluate on LongVideoBench (Wu et al., 2024) (the complete official validation set, $n=1337$), Video-MME (Fu et al., 2025) ($n=2700$), and LVBench (Wang et al., 2024) ($n=1549$). Tables abbreviate the first two as LVB and V-MME where column width requires it. LongVideoBench is additionally stratified into 15, 60, 600, and 3600s duration bins, which is what makes the duration analysis in §5.1 possible. The primary answerer is Qwen3-VL-8B-Instruct (Qwen Team, 2025), run through lmms-eval (EvolvingLMs-Lab, 2024) with greedy decoding, temperature zero,

and subtitles disabled. InternVL3-2B and InternVL3-8B (Zhu et al., 2025a) provide a second open-model family, and GPT-5-mini checks selected contrasts on the same saved timestamps.

Statistics. Accuracy comparisons use exact two-sided McNemar tests (McNemar, 1947) on paired correctness outcomes, which is the appropriate test when the same question is answered under both policies. Compression claims need more than a non-significant McNemar result, because failing to detect a difference is not evidence that none exists. Where we claim two arms are interchangeable we therefore report two one-sided tests (TOST) with 90% confidence intervals (Schuirmann, 1987), which bound the effect rather than merely failing to find it. The ± 2 , ± 3 , and ± 4 percentage-point margins were chosen after seeing the runs rather than preregistered; we therefore report both the narrowest margin the data support and the adjacent margin they do not. Tests are contrast-specific and uncorrected for multiplicity.

5 Results

5.1 Choosing frames beats having more of them

The most direct evidence that selection matters is that it substitutes for frame count. On 3600s LongVideoBench videos, OMP with eight frames reaches .5461 while uniform sampling with sixteen frames reaches .4770—a 6.9-point gain on half the input ($p=.0011$). The same pattern holds one budget higher: on 600s videos, OMP with sixteen frames reaches .6578 against .6044 for uniform sampling with thirty-two, a 5.3-point gain at half the frame count ($p=.018$). In both, the policy that is cheaper in answerer visual tokens wins—a stronger statement than any equal-budget comparison can make, though the accounting covers the answerer only: producing the selected set also requires decoding at 1 fps and encoding the whole pool with LongCLIP, a cost we do not measure and which a single-query-per-video deployment would not amortise. Doubling the frame count is the obvious way to spend more compute on a long video, and on these benchmarks it is the worse way.

Table 1 places that result in the wider comparison, with all six rules reading the same cached scores. OMP improves on uniform sampling by 5.69 points on LongVideoBench, 5.85 on Video-MME, and 11.81 on LVBench, and the paired evidence is decisive on Video-MME ($p=5.1 \times 10^{-10}$), LVBench ($p=9.5 \times 10^{-17}$), and the LongVideoBench 3600s bin ($p=5.9 \times 10^{-4}$). LDDR-select is the one rule that keeps pace, staying within a point of OMP on all three benchmarks with no detectable paired difference on Video-MME or LVBench ($p=.70$ and $.57$).

The duration split explains when any of this matters. On 15s clips, uniform sampling, top- k , and OMP return exactly the same accuracy (.7249): eight frames out of fifteen candidates cover the video regardless of how they are picked, so there is nothing for a selector to do. The gains appear once the candidate pool outgrows the budget—7.8 points at 600s ($p=.0022$) and 7.5 points at 3600s ($p=5.9 \times 10^{-4}$). Selection is a scarcity effect, and the field’s interest in it should scale with video length rather than apply uniformly.

Which kind of rule is needed is more benchmark-dependent than the headline suggests. On Video-MME, top- k , AKS, and FOCUS* do not separate detectably from uniform sampling, while OMP and LDDR-select do; on LongVideoBench, OMP’s advantage over plain top- k is small and often undetectable within individual bins. The supported reading is narrow and worth stating precisely: combining query relevance with suppression of already-covered directions is consistently strong in this harness, whereas the value of any individual simpler heuristic does not transfer across benchmarks on its own. Whether that ordering is a property of the rules or of LongCLIP is the subject of §5.2.

5.2 The ranking survives a scorer swap

Every number in Table 1 comes from a single frozen scorer, which leaves open whether the ordering reflects the selection rules or LongCLIP’s particular geometry. We test that directly, replacing LongCLIP with SigLIP-so400m (Zhai et al., 2023) and holding the prompt boundary, frame budget, answerer, and harness fixed. Both scorers’ arms were re-run together in one environment rather than

Method	LongVideoBench	Video-MME	LVBench
Uniform	.5654 (−5.69)	.5637 (−5.85)	.3454 (−11.81)
Top- k	.6028 (−1.95)	.5704 (−5.18)	.4319 (−3.16)
OMP	.6223	.6222	.4635
AKS	.6021 (−2.02)	.5785 (−4.37)	.4280 (−3.55)
FOCUS*	.5819 (−4.04)	.5578 (−6.44)	.3983 (−6.52)
LDDR-select	.6320 (+0.97)	.6193 (−0.29)	.4693 (+0.58)

Table 1: Matched selector comparison at $k=8$ with Qwen3-VL-8B. Cells are accuracy; parentheses give the difference from OMP in percentage points. FOCUS* is the controlled schedule replay described in Appendix B; LDDR-select is stage 1 only.

comparing fresh SigLIP numbers against the banked LongCLIP results, so every contrast below is paired on identical items.

The swap is substantial rather than cosmetic: on the 600s bin the scorers agree on only 1.32 of eight frames for OMP and 2.62 for top- k —seven to fourteen times chance for these pool sizes, so both are selecting rather than scattering, yet 67 to 84% of what the answerer sees changes.

Table 4 shows the ordering surviving intact. Uniform sampling is beaten by top- k and top- k by OMP under both scorers, and OMP’s advantage over uniform is significant in each case ($p=.0024$ under LongCLIP, $p=.00026$ under SigLIP). Discordant counts for these contrasts are $b=63, c=41$ and $b=69, c=37$ under LongCLIP and $b=60, c=38$ and $b=65, c=29$ under SigLIP, for top- k and OMP respectively. Neither selector’s gain depends detectably on which scorer produced it, but the test is not powerful: with roughly 134 discordant pairs the 90% interval on the OMP scorer effect spans about $[-3.6, +5.6]$ points, so this bounds large scorer effects rather than establishing their absence. The 60s bin adds nothing either way: no contrast there reaches significance (the largest is $+3.49$ points at $p=.31$), which is the expected behaviour when eight frames already cover 22% of a median 36-frame pool.

One reading must be resisted. OMP beats top- k significantly under SigLIP ($+3.40$, $p=.049$) and not under LongCLIP ($+2.43$, $p=.33$), which invites the conclusion that OMP suits SigLIP better. The interaction test rejects that: the difference of the two gains is 0.97 points ($b=48, c=43, p=.68$). That estimate coincides with the OMP row of Table 4 only because top- k has identical marginal accuracy under both scorers; the two are dif-

ferent tests over different per-item patterns, which is why their p -values differ. A significant result beside a non-significant one is not evidence that the two differ, and Appendix C records the same error made on question categories.

The claim is bounded: one alternative scorer on two duration bins does not establish scorer-invariance, and both encoders share a contrastive image-text objective, so an ITM head or an attention-based scorer could still reorder the table. What it rules out is the specific worry that Table 1 is an artifact of LongCLIP—the ordering holds when most selected frames are replaced.

5.3 Half the pixels change almost nothing

Table 2 holds the selected timestamps fixed and changes only how many pixels each frame is worth. Across the three Qwen benchmark aggregates, moving to the compressed arm shifts accuracy by at most 0.44 points in either direction. A uniform-timestamp control on pooled LongVideoBench long videos moves by 0.31 points ($p=.784$), which shows the stability is a property of the compression, not something special about OMP-selected frames.

Small differences are easy to over-read, so we bound one of them properly. On the pooled LongVideoBench long bins ($n=976$), the compressed OMP arm is 0.72 points higher than full resolution, with a 90% confidence interval of $[-0.60, +2.03]$. That interval fits inside a ± 3 -point margin (TOST $p=.0022$) but narrowly fails at ± 2 ($p=.054$), so ± 3 is the honest statement. InternVL3-8B supports a stronger version under a more aggressive intervention: cutting 24 tiles to 8 moves pooled accuracy by -0.10 points with a 90% interval of $[-1.66, +1.45]$, establishing equivalence at ± 2 points ($p=.022$). The remaining LVBench

(a) Fixed timestamps, halved spatial budget			
Answerer / benchmark	Full	Compressed	Δ (pp)
Qwen / LongVideoBench	.6223	.6253	+0.30
Qwen / Video-MME	.6222	.6178	-0.44
Qwen / LVBench	.4635	.4674	+0.39
GPT-5-mini / LVBench	.4900	.4926	+0.26 ($p=.84$)
Uniform / LVB long pool	.5061	.5092	+0.31 ($p=.784$)

(b) Reinvestment at equal or lower token cost			
Answerer / benchmark	$k=8$ full	$k=16$ compr.	Δ (pp)
Qwen / LongVideoBench	.6223	.6447	+2.24
Qwen / LVB long pool	.5820	.6055	+2.36 ($p=.0346$)
Qwen / Video-MME	.6222	.6378	+1.56
Qwen / LVBench	.4635	.4939	+3.04 ($p=.0009$)
GPT-5-mini / LVBench	.4900	.5139	+2.39 ($p=.039$)

Table 2: (a) Spatial compression at fixed timestamps. Qwen rows use the approximately half-budget arm; the GPT row uses a pixel-ratio control, and the uniform control an approximately 46% realized per-frame budget. (b) Reinvestment; token parity is audited on the LongVideoBench bins only, and Video-MME and LVBench use the same resize rule without a separate audit.

and Video-MME differences in Table 2a are descriptive stability checks; we do not present them as equivalence results.

Compression being cheap does not mean the allocation shape is irrelevant, and our controls separate the two. The residual-proportional schedule does not beat a flat split at the same mean budget (Figure 3), so OMP’s ranking—useful as it is for choosing frames—carries no usable signal about how to distribute pixels among them. A reconstruction of LDDR’s stage-2 Group-DPP importance does better, improving on the flat split by 1.84 points on the pooled long bins ($p=.0198$, 95% CI $[+0.37, +3.32]$). That contrast comes from our own same-environment reconstruction rather than the authors’ released implementation, and its 3600s component is not individually significant, so we treat it as indicative. The conclusion is therefore three-part and narrower than “resolution does not matter”: coarse compression is close to free, a naive priority-proportional rule adds nothing, and a better-designed importance signal can add something.

5.4 Spending the savings on more frames

The direction of this intervention is not new: Chen et al. (2026a) report fixed-resolution ablations under one total token budget where 1024 tokens per frame reach 59.76 on LongVideoBench, 512 reach 61.03, and 256 fall back to 59.31, so halving the per-frame budget doubles the frames that fit and

accuracy improves by 1.27 points before declining again. That ablation does not establish whether the parity it assumes holds, whether the difference exceeds noise, or whether the effect needs their selector; we address those three, making this an audited and paired replication rather than a new observation. Table 2b spends it on temporal coverage, comparing eight full-resolution frames against sixteen compressed ones at equal or lower measured token cost. LongVideoBench improves by 2.24 points overall and 2.36 on the pooled 600 and 3600s bins ($p=.0346$), with almost identical positive effects in each. LVBench gains 3.04 points ($p=.0009$), Video-MME 1.56, and GPT-5-mini on LVBench 2.39 ($p=.039$).

We cannot show that this requires OMP. All four cells of the selector-by-budget design—uniform and OMP, each at eight full-resolution and sixteen compressed frames—were run in a single environment, so the interaction involves no cross-environment comparison. Uniform gives +1.13 points ($p=.3673$) on the eight-to-sixteen move; OMP gives +2.56. The interaction between them is 1.43 points and does not reach significance ($p=.2614$), with OMP gaining on 110 items where uniform does not and uniform gaining on 93 where OMP does not. It does split by duration—essentially zero at 600s ($+0.00$, $p=1.000$) and +2.48 at 3600s ($p=.1706$), which is the direction expected if selection matters more as the candidate pool grows—but neither bin is individually significant. This is evidence against a large OMP-specific interaction rather than proof the two are identical; with 203 discordant pairs the test excludes only large effects. The defensible claim is that temporal reinvestment works with OMP, points the same way under uniform sampling, and that its selector-specificity remains open at roughly the one-point scale.

5.5 What the later picks actually add

OMP’s residual trace does not behave the way its signal-processing origin predicts, and the discrepancy explains why extra frames still help. On both LongVideoBench long bins the residual norm is 0.972 of its starting value after one pick and 0.967 after eight, falling no further by sixteen: almost none of the query is ever reconstructed. What does change is the residual’s correlation with the selected frame,

dropping from .233 at pick one to .004 at pick eight and roughly zero by sixteen, so the query-directed signal is exhausted early. Yet those later frames keep a cosine near .20 with the original query.

The text query sits almost orthogonal to the dominant directions of the frame-embedding cone—the modality gap of contrastive image-text encoders (Liang et al., 2022), visible in Figure 1b—so with an effectively low-rank frame dictionary, eight greedy picks reach only those directions and OMP cannot reduce the residual much whatever it selects. What it can do is avoid repeating itself: late picks add little new query direction while adding chances for the answerer to encounter a relevant moment, which is what the reinvestment result rewards.

Coverage helps only while it stays relevant. An exploratory inspection of 93 OMP failures at $k=8$ across the two long bins found one dominant pattern—off-topic novelty, in 20 of 41 inspected 600s cases and 35 of 52 inspected 3600s cases, where picks went to visually distinctive but question-irrelevant material. The mirror-image failure also appeared: on cross-scene tracking questions, dense top- k clustered on one relevant moment and missed another required scene. Category-level $k=8$ versus $k=16$ results are mixed, so this supports a mechanism hypothesis rather than a rule about question types, but it points at a design target: a relevance floor, so spread is explored within plausible evidence regions.

5.6 The same frames do not help every model equally

Table 5 hands the identical OMP and uniform timestamps to InternVL3, changing the answerer and nothing else. The selection gain replicates on LongVideoBench and LVBench at both model sizes. Video-MME does not cooperate: the 2B model benefits at every duration, while the 8B model benefits only on short videos and is flat on medium and long ones despite receiving demonstrably different frames. Model capacity is not the explanation, because the same 8B model gains 10.14 points on LVBench.

The allocation results transfer more smoothly. On pooled LongVideoBench long videos, InternVL3-8B holds its accuracy through a threefold tile reduction (−0.10

points, $p=1.000$) and gains 3.38 points when the same total tile budget is spread from 8 uniformly sampled frames to 24 ($p=.0099$), an effect concentrated in the 600s bin. Both halves of the select-compress-reinvest story therefore survive a change of answerer, but their magnitudes do not: selection and allocation effects are properties of an answerer-benchmark pair, not constants of a timestamp set.

6 Discussion

The three decisions are not equally worth optimizing. Under a fixed visual-token budget the interventions rank consistently. Choosing frames is the large lever, sometimes beating uniform inputs that carry twice as many frames. Compressing those frames is close to free but buys nothing alone—it changes few predictions in every Qwen and InternVL setting we tested—so its value lies entirely in what the savings buy, and temporal coverage returns a further two to three points. Fix selection first, treat resolution as the slack variable, and reinvest rather than pocket the difference.

A 1993 algorithm is competitive with purpose-built selectors. The most uncomfortable observation in Table 1 is which rules sit close together. Orthogonal Matching Pursuit predates every method it is compared against by three decades, has no video-specific machinery and no tuned hyperparameters, and still leads AKS, FOCUS*, and top- k on all three benchmarks while staying within a point of LDDR’s stage-1 selector. This is not evidence that recent selectors are poorly designed; it is evidence about where their gains live. Once the scorer, prompt boundary, frame budget, and answerer are held fixed, the room between subset-selection rules is usually modest: on LongVideoBench and LVBench it is smaller than the distance from uniform sampling to plain top- k . Video-MME is the exception, and an instructive one. There top- k gains only 0.67 points over uniform while OMP and LDDR-select gain close to six, so the choice of which rule matters far more than the choice to be query-aware at all. The defensible generalization is therefore narrower than “subset rules barely matter”: on two of three benchmarks

the rule contributes less than the scorer and pipeline it arrives with, and on the third the rule itself is decisive. The corollary for evaluation practice is concrete: a selector comparison is only informative if the scoring stage is shared, and a paper reporting a new rule should include a classical pursuit or greedy baseline on its own scores before claiming the rule is what helped.

Two matched harnesses disagree by as much as the effects they measure. Appendix A sets our gains beside the one prior study that standardises its encoder, at a cell matched on answerer family, budget, and encoder. The two harnesses differ by 0.4 to 3.8 points on identical rules—a range covering most of the gaps published selector comparisons are built on—which is why every claim here is a paired contrast inside one harness rather than a comparison against a published figure.

Joint allocation systems deserve decomposed reporting. This decomposition sharpens how to read systems such as Q-Frame, LDDR, and DAFS (Zhang et al., 2025a; Chen et al., 2026a; Wang et al., 2026b), whose end-to-end gains combine several causal routes. Our controls find no value in a priority-proportional resolution rule, a positive effect from reconstructed LDDR-style importance, and the clearest downstream gain from temporal reinvestment—three answers a single end-to-end number would have blurred into one. Future methods would be easier to build on if they reported separate timestamp, resolution, and frame-count interventions alongside the joint result.

The mechanism suggests where to look next. The text query is nearly orthogonal to the dominant directions of the frame-embedding cone (Liang et al., 2022), so the residual barely shrinks and sparse reconstruction never really happens; what remains useful is relevance plus decorrelation. Late picks extend multi-scene coverage after the text residual flattens, but unconstrained diversity also rewards irrelevant novelty, which our failure audit identifies as the dominant OMP-specific error. Relevance floors, event-boundary constraints, and scorers with a smaller modality gap are the natural next interventions.

Conclusions should span durations and answerers. Selection is a property of the answerer–benchmark pair: InternVL3-8B is nearly insensitive to OMP on medium and long Video-MME yet gains over ten points on LVBench, and OMP’s advantage is exactly zero at 15s. A claim measured on one benchmark at one duration with one answerer is not safe to generalize, and our own results would have supported at least three different headlines depending on which cell we had chosen to report.

7 Limitations

Two selectors are controlled reproductions, not full systems. FOCUS* replays the published selection schedule against dense LongCLIP scores instead of reproducing the original budgeted ITM scorer, so its row tests the schedule under a matched scorer and not the FOCUS pipeline’s accuracy or efficiency. LDDR-select is stage 1 only, and the stage-2 allocation contrast is reconstructed from the paper rather than run from the authors’ code. Neither row should be read as a reproduction result for the original method.

Equivalence is established in one primary setting. The formal Qwen equivalence result covers pooled LongVideoBench 600 and 3600s examples within a post-hoc ± 3 -point margin, and misses ± 2 . Video-MME compression has no paired test, and the small LVBench differences are descriptive rather than equivalence evidence. The selector-by-budget interaction is underpowered for effects near one point, which is the scale at which it would matter.

Mechanism evidence is encoder-specific. The residual geometry and Gram statistics come from frozen LongCLIP stem embeddings, mostly on LongVideoBench-600s. The selector ordering survives a swap to SigLIP (§5.2), but both encoders share a contrastive image–text objective; BLIP-ITM heads, MLLM-attention scorers, and subtitle-aware selection remain untested, and the modality-gap explanation in §5.5 is a property of this embedding space rather than of pursuit selection in general. The model-assisted failure audit examined selected rather than randomly sampled failures, so its counts are exploratory and can-

not estimate population rates.

Some secondary results were measured in a second compute environment. Tables 1–?? and every claim drawn from them come from a single stack: RTX PRO 4500 GPUs with PyTorch 2.11/CUDA 13.0. The selector-by-budget interaction in §5.4, the allocation-shape contrast in §5.3, and the variant sweep in Appendix D were measured on L40S GPUs with PyTorch 2.6. In each case every arm of the contrast—including a re-run of the full-resolution baseline—was executed inside that second environment, so no reported difference is computed across the two stacks. Replicating the baseline in both also lets us quantify the drift rather than assume it is small: the uniform eight-frame full-resolution arm reads 0.36 points higher on the second stack in the 3600s bin and is identical in the 600s bin, while the corresponding OMP arm reads 0.71 points lower. Absolute accuracies from these sections are therefore not comparable to Tables 1–?? and we never place them in the same table; the paired differences within each section are unaffected.

Evaluation scope is incomplete. Subtitles are disabled throughout, which leaves subtitle-anchored questions partly inaccessible to every visual selector and depresses all arms together. GPT-5-mini compression uses a pixel ratio rather than Qwen token accounting, and some GPT selector results survive only as aggregates rather than per-item records.

8 Conclusion

Under matched controls, query-aware keyframe selection is the dominant decision in long-video question answering. Eight well-chosen frames beat sixteen uniformly spaced ones on hour-long videos, and a greedy pursuit algorithm from 1993 is enough to get that result—outperforming uniform sampling by 5.7 to 11.8 points and staying within one point of a purpose-built modern selector. Roughly half of the per-frame spatial budget can then be removed with little observed change, with ± 3 -point equivalence on pooled LongVideoBench long videos, and reinvesting those tokens in additional keyframes returns a further two to three points. Select relevant

frames first, compress them second, and use the savings to see more of the video—while remembering that how much any of this helps depends on the video’s length, the benchmark, and the model doing the reading.

9 Ethical Considerations

The experiments use existing public video-QA benchmarks and frozen open or API models. No new human subjects, personal data, videos, or annotations were collected. Question stems, but not answer options, enter the selector. Benchmark videos remain governed by their original licenses and are not redistributed with the released artifacts.

10 Data and Code Availability

Evaluation code, selector implementations, analysis scripts, per-item predictions permitted by the benchmark licenses, and aggregate result tables are available at <https://github.com/codeprakhar25/omp-keyframe-sampling>. The release includes the scorer-swap pipeline of §5.2, the fused-query arms of §3, and the script that builds Table 3 from the cited papers’ reported numbers. The release excludes benchmark video files and API credentials.

11 Author Contributions

Prakhar Khatri: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing—Original Draft, and Writing—Review & Editing.

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13 Competing Interests

The author declares no competing interests.

14 AI Assistance Disclosure

Generative AI tools assisted code development, manuscript editing, and an exploratory visual failure audit. The author directed the analyses, verified the reported numerical results against saved artifacts, adjudicated the scientific claims, and takes responsibility for the final manuscript.

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A Relocated tables and figures

Two matched harnesses disagree by as much as the effects they measure. LDDR is the one prior study that standardises every baseline to a single encoder, and it reports a cell matched to ours on answerer family, frame budget, and encoder. Table 3 places that cell beside our own. Our gains are smaller in eight of the nine comparable cells, by 0.4 to 3.8 points. We do not read this as either harness being wrong. The uniform baselines themselves differ—most starkly on LVBench, 28.08 against our 34.54—so the two harnesses are not measuring from the same floor, and a lower floor leaves more

Rule	Video-MME		LongVideoBench		LVBench	
	pub.	ours	pub.	ours	pub.	ours
uniform (absolute)	57.19	56.37	54.53	56.54	28.08	34.54
AKS [†]	+4.29	+1.48	+5.01	+3.67	+12.07	+8.26
FOCUS ^{*†}	+1.70	−0.59	+5.00	+1.65	+3.49	+5.29
LDDR-select	+5.96	+5.56	+8.90	+6.66	+14.98	+12.39
OMP	—	+5.85	—	+5.69	—	+11.81

Table 3: Two matched-condition harnesses, same rules. Published columns are [Chen et al. \(2026a\)](#), Table 1, Qwen3-VL-8B at eight frames with a LongCLIP encoder—the same answerer family, budget, and encoder as ours—where LDDR-select is their stage-1 row. The first line gives absolute uniform-sampling accuracy; all other cells are gains over the uniform row of the same column. [†] marks rules we run on a substituted scorer (both score with BLIP-ITM natively); ^{*} marks the FOCUS schedule replay of Appendix B. OMP has no published counterpart. MDP3 and Q-Frame appear in their table and we did not run them.

Selector	LongCLIP	SigLIP	SigLIP − LongCLIP
Uniform	.5534	—	—
Top- <i>k</i>	.6068	.6068	+0.00 (<i>p</i> =1.00)
OMP	.6311	.6408	+0.97 (<i>p</i> =.73)

Table 4: Scorer swap on LongVideoBench-600s (*n*=412), all arms run together and paired on identical items. Uniform sampling is scorer-independent and appears once. The final column is a paired test of whether each selector’s gain depends on the scorer.

room for a selector to gain. Prompt template, candidate-pool construction, decoding path, and answer parsing all differ and none of them is specified tightly enough in either paper for a reader to reconcile the two. We also find no clean pattern in the residuals: the rule whose native encoder we did not substitute (LDDR-select, −0.40 on Video-MME) reproduces closest there but not on LongVideoBench, where AKS is closer despite a substituted scorer.

The disagreement is itself the measurement. Two carefully controlled harnesses, running the same published rules at the same budget with the same encoder, differ by 0.4 to 3.8 points—a range that covers most of the gaps published selector comparisons are built on. That bounds how much weight any cross-paper selector number can carry, including the numbers this paper’s own argument targets, and it is why every claim we make is a paired contrast inside one harness rather than a comparison against a published figure. It

The dictionary is coherent and the query is near-orthogonal to it, so pick 1 removes 85.6% of all query mass OMP ever removes

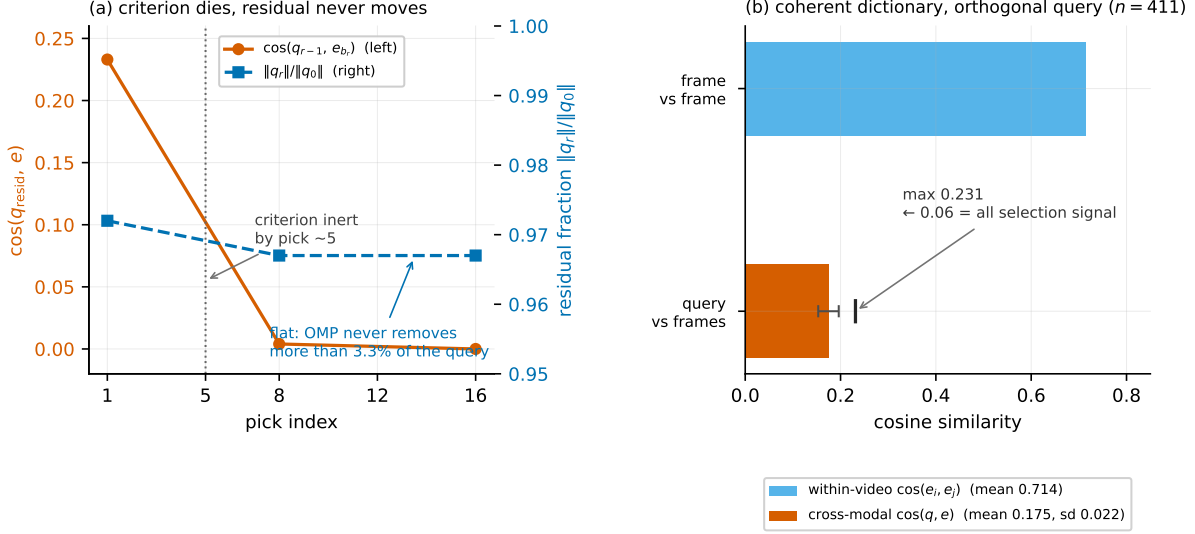


Figure 1: LongCLIP geometry on LongVideoBench-600s ($n=411$). (a) Residual correlation vanishes while residual norm changes little; the line connects logged picks 1, 8, and 16 and is not a dense trace. (b) Within-video frames occupy a coherent cone, while the text query is nearly orthogonal to the frame dictionary. In this space, OMP mainly suppresses directions already represented by earlier selections.

Benchmark	n	IVL3-2B Δ	IVL3-8B Δ
Video-MME short	900	+6.00 ($p=1.2 \times 10^{-4}$)	+5.44 ($p=1.9 \times 10^{-4}$)
Video-MME medium	900	+3.33 ($p=.046$)	+0.33 ($p=.891$)
Video-MME long	900	+4.22 ($p=.0062$)	+0.11 ($p=1.000$)
LVBench	1549	+9.04 ($p=7.2 \times 10^{-11}$)	+10.14 ($p=1.2 \times 10^{-13}$)
LongVideoBench	1337	+4.71 ($p=1.6 \times 10^{-4}$)	+6.51 ($p<10^{-4}$)

Table 5: OMP minus uniform accuracy on identical timestamps and tile budgets with InternVL3. Deltas are percentage points.

also means Table 3 should be read as a measurement of harness disagreement, not as a reproduction verdict on any of these methods.

Benchmarks	LongVideoBench (1337), Video-MME (2700), LVBench (1549)
Primary answerer	Qwen3-VL-8B, greedy decoding, subtitles off
Transfer checks	InternVL3 2B/8B; GPT-5-mini
Shared scorer	LongCLIP, question stem only, 1 fps pool
Scorer control	SigLIP-so400m, identical protocol (§5.2)
Frame budgets	$k=8$ for selection; $k=16$ for reinvestment

Table 6: Evaluation grid. Main claims concern paired accuracy differences under matched inputs, not absolute leaderboard position.

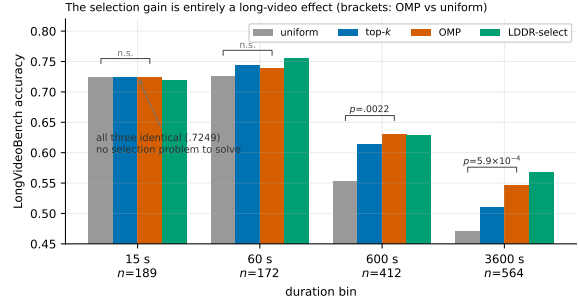


Figure 2: OMP’s gain over uniform sampling grows with video duration on LongVideoBench. At 15s, uniform, top- k , and OMP have identical accuracy (.7249); the significant gains occur at 600 and 3600s.

B FOCUS* schedule replay

Every row in Table 1, including FOCUS*, consumes the same dense 1 fps LongCLIP stem-score vector. The original FOCUS method instead treats scoring itself as a budgeted on-line process, in which clip-arm pulls purchase ITM evidence (Zhu et al., 2025b). Our replay preserves the temporal clip schedule and default hyperparameters but reveals already-computed LongCLIP scores to the bandit. It therefore tests the schedule under a matched scorer, and says nothing about the efficiency

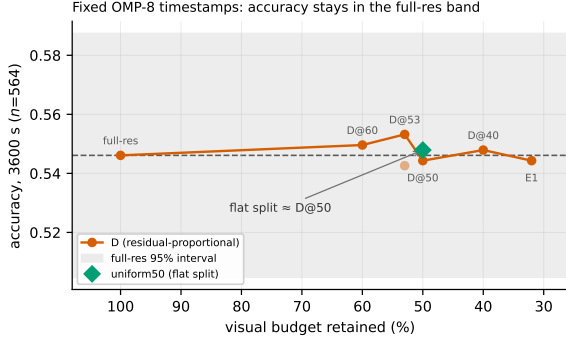


Figure 3: LongVideoBench 3600s accuracy on fixed OMP-8 timestamps across spatial budgets. The flat 50% control and the residual-proportional schedule are indistinguishable, so OMP rank alone does not provide a useful resolution policy.

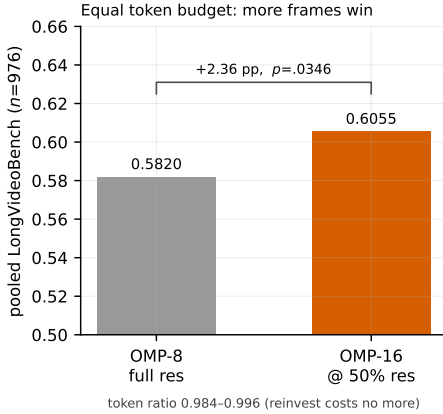


Figure 4: LongVideoBench 600s and 3600s pooled: sixteen compressed OMP frames outperform eight full-resolution OMP frames at a measured token ratio no greater than one.

or accuracy of the full FOCUS pipeline.

The replay uses 16s clips, three coarse pulls per clip, $\alpha=0.25$, a pull budget equal to half the candidate pool, and approximately $k/4$ surviving clips. Unobserved rewards are interpolated from the nearest observed frame, and the random seed is fixed per item. This boundary is why the row carries a star, and why we draw no conclusion from it about FOCUS under its native ITM protocol.

C Question categories do not define a stable selector regime

LongVideoBench provides question-category tags, which invite the hypothesis that temporally referred questions need different frame selection from the rest. We test it by comparing OMP with uniform sampling on the

temporally referred categories (the T* group) against all remaining categories, without relabeling the data.

Bin	Slice	n	Uniform	OMP	Δ (McNemar)
600s	temporal T*	148	.5608	.5811	+2.03, $p=.74$
600s	other	264	.5492	.6591	+10.98, $p=5.2 \times 10^{-4}$
3600s	temporal T*	222	.4369	.5270	+9.01, $p=.013$
3600s	other	342	.4942	.5585	+6.43, $p=.021$

Table 7: OMP minus uniform on official LongVideoBench categories. Percentage-point gains differ by duration, and the category interaction is not significant in either long bin.

The temporal slice looks resistant to selection at 600s and responsive at 3600s, which is the shape of a result that would be easy to over-interpret. It is not one. The category interaction is not significant in either bin, and which slice “wins” reverses between them. Many T* stems refer to subtitles or speech while our selector is vision-only and subtitles are disabled, which offers one plausible source of the instability. Two significant-versus-non-significant subgroup tests are not an interaction, so we infer no question-type router from these results.

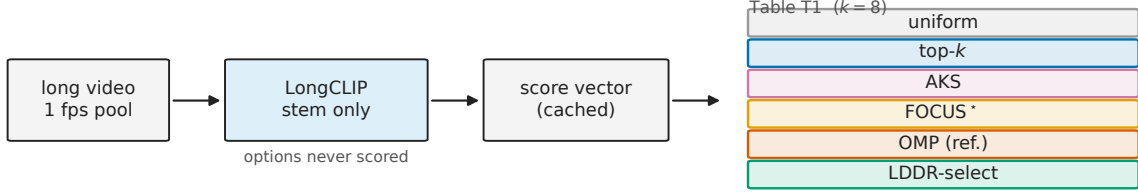
D Selector variants

This sweep changes only the subset rule, on the same LongCLIP cache. MMR underperforms; query-blind DPP ($\beta=0$) collapses toward uniform sampling, which confirms that the diversity term carries no signal on its own; and query-weighted DPP, residual floors, and partial orthogonalization all sit in a band around OMP. No tested DPP variant significantly exceeds OMP in either long bin.

This sweep ran in the secondary compute environment described in §7; its arms are internally paired against each other and are never compared against Tables 1–??.

These nulls do not show that all diversity objectives are equivalent. They show something more specific and more useful: within this scorer geometry and budget range, moving around inside the relevance–diversity family produces less change than replacing uniform sampling or reallocating pixels into frames. That is consistent with the residual analysis in §5.5—if the query-directed signal is exhausted after roughly five picks, then rules that differ only in how they trade relevance

(a) Score once. Every rule reads the same LongCLIP stem vector.



(b) Then spend the same tokens two ways (timestamps frozen).

$k = 8$ full resolution

$k = 16$ at $\sim 50\%$ resolution

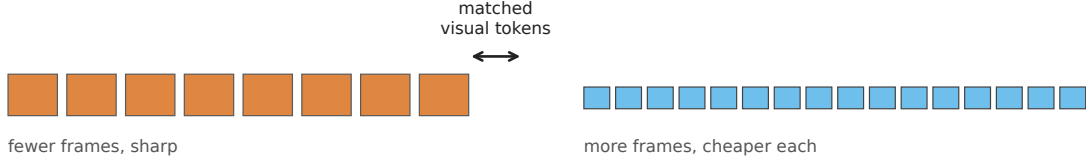


Figure 5: Controlled protocol. (a) A 1 fps candidate pool is encoded once with LongCLIP using only the question stem; every $k=8$ selector consumes the same cached embeddings and scores. (b) Selected timestamps are then frozen while spatial budget is reduced. (c) The saved visual tokens are reinvested by comparing eight full-resolution frames with sixteen compressed frames at equal or lower measured token cost. FOCUS* denotes a replay of the published selection schedule on dense LongCLIP scores, not its original budgeted ITM scorer.

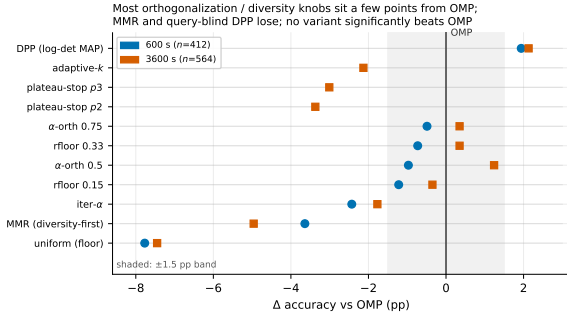


Figure 6: Coverage-gated selector variants against OMP at $k=8$ on the 600 and 3600s LongVideoBench bins. The shaded ± 1.5 -point band is a visual aid, not a statistical criterion.

against diversity have little left to differ about.

E Exploratory failure audit

Two independent model-assisted visual passes inspected OMP failures at $k=8$ through a frame viewer: 41 cases spanning 16 categories in the 600s bin, and 52 cases spanning 17 categories in the 3600s bin. Cases were chosen to cover categories and to include both OMP-only and shared failures; the sample is not random. Reviewers compared the selected timestamps, the rendered frames, the question, the gold answer, and the model’s predic-

tion. Primary tags were missed evidence, temporal offset, answerer error despite adequate frames, and questions not answerable from visual frames at all.

Both passes converged on the same dominant OMP-specific pattern: visually distinctive but irrelevant content consuming several of the eight picks—unrelated chapters in compilations, separate storylines in long narrative video, title and subscribe cards, and dark transitions. The opposite failure showed up on sequence-of-scenes and cross-scene tracking questions, where dense top- k sampling covered one relevant moment but omitted another required scene. The audit therefore motivates relevance-constrained diversity as a design direction. It cannot estimate population failure rates, and it cannot establish that adding frames benefits any particular question category.